Introduction to Computational Methods for Digital Humanists

VU MA Digitale Geisteswissenschaften, University of Graz, Summer term 2027, Wednesdays, 10:15 – 11:45 A.M.

Session	Title	Content and Learning Objectives	Preparation for the session
Session 1 03.03.27	**Welcome & organization**	Organization: content, learning objectives, Moodle; the learning environment and your peers. Brainstorm research ideas: what data do you work with and what data do you want to work with? Let's look at text, image, audio, and video data Overview of the computational-methods pipeline in digital humanities: corpus, model, annotate, analyze, evaluate, reflect, disseminate.	Reading: Semester welcome letter Check: Minimal technical requirements for the class
Session 2 10.03.27	**Corpus Building I: Corpora in the computational humanities**	What is a corpus? What is a subcorpus? Which kinds of corpora exist? Where to find existing corpora for your computational humanities research area: text corpora and image and audio corpora, for example an IIIF image collections, GLAM and museum APIs, and oral-history archives Copyright and FAIR data; text-and-data-mining exceptions; derived text formats.	Reading: Bläß (2024[2020]), "Korpusbildung" (forTEXT journal; "Corpus Building" English translation on request). Schöch (2026), "Derived Text Formats" Instructional reading: How to work in groups Group work: find two to three partners and choose a group on Moodle and formulate your group-work agreement
Session 3 17.03.27	**Corpus Building II: sources, rights, acquisition**	Formulate research questions and hypotheses. Corpus design, file types, and documentation. Acquiring text: the Wikipedia API, Project Gutenberg, IIIF endpoint; what may and may not be redistributed (CC BY-SA, public domain, copyright).	Instructional reading: How to prepare the portfolio Read and try: Milligan, "Automated Downloading with Wget" (Programming Historian) + Jupyter notebook fetch_wikipedia_tei.ipynb Brainstorm: which corpus would you like to build? Group work: start building your corpus and define your research question
—	**Time off**	Public holiday / semester break (22.03.-03.04.2027)	
Session 4 07.04.27	**Corpus Building III: metadata and text preparation**	From raw text to structured text (known from TEI P5 basics in "X-Technologien I"). Text preparation pipeline: from HTML to TEI. Metadata enrichment: beyond the metadata from the HTML (Metadata standards such as Dublin Core or MODS, known from "Informationsmodellierung")	Reading: Drucker (2021), DH Coursebook, ch. 4 "Metadata, markup, and data description" Read and try: Momen et al. (2025) "Filling the temporal void" + Jupyter notebook metadata_enrichment.ipynb Group work: continue building your corpus
13.04.27	**Deadline: Portfolio Part 1**	Submit your first portfolio chapter (research question and corpus description) as a PDF on Moodle.	
Session 5 14.04.27	**Data Preparation: Data cleaning, tokenization, sentence splitting**	Cleaning and preparing text for computational analysis. Word tokenization, sentence segmentation, regular expressions, etc.	Reading: Jurafsky/Martin (2026), Speech and Language Processing, ch. 2 "Words and tokens" Group work: add metadata to your corpus

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Session 6 21.04.27	**Annotation I: Named entity recognition and object detection**	What is annotation? Named entity recognition for text data, focus on entity types (PER, ORG, LOC, GPE). Beyond text data: visual analogue in object detection; Annotation as interpretation; theory pluralism and the guideline-dependence of categories. Intersubjectivity as the goal of annotation.	Reading: Ehrmann et al. (2023) "Named Entity Recognition and Classification in Historical Documents: A Survey" Group work: brainstorm whether named entity recognition is a useful method for your research question; experiment with the text cleaning and preparation steps (Jupyter notebooks)
Session 7 28.04.27	**Annotation II: Annotation-based operationalization**	How to get from a humanities phenomenon to its operationalization for computational text analysis. Introduction to systematic manual annotation.	Reading: Pichler/Reiter (2022), "From Concepts to Texts and Back: Operationalization as a Core Activity of Digital Humanities" Group work: brainstorm about the phenomenon you want to analyze to answer your research question
Session 8 05.05.27	**Annotation III: Formulating intersubjective annotation guidelines, Evaluation**	Iterative preparation of annotation guidelines: Tagset creation and guideline formulation. What are compulsory and facultative parts of annotation guidelines? Inter-annotator agreement measurements	Reading: Reiter & Guhr (2025), "Instructions for Creating Annotation Guidelines" Reading (for repetition): Andresen & Zinsmeister (2019), Korpuslinguistik (excerpt on "Corpus annotation"; German original or English translation provided on request). Group work: brainstorm about useful annotation for approaching your research question
Session 9 12.05.27	**Annotation IV: Automated annotation with transformer models, evaluation**	Operationalization from phenomenon to automated annotation pipeline: training and test data preparation, transformer model family (text (BERT), vision (vision transformers) and speech (wav2vec, Whisper); Evaluating automatic annotation: precision, recall, F1, accuracy	Reading: Jurafski/Martin (2026), Speech and Language Processing, ch. 7 "Transformers and Pretraining" (optional) Reading: Vaswani et al. (2023[2017]), "Attention is all you need" Group work: formulate your annotation guidelines and prepare some manual annotations on your corpus data
Session 10 19.05.27	**Annotation V: Distant reading vs. distant viewing**	On the methodological and theoretical framework for the study of large collections of visual materials	Reading: Arnold/Tilton (2019), "Distant viewing: analyzing large visual corpora" Group work: brainstorm about multimodal approaches to your research question
Session 11 26.05.27	**Annotation VI: Large language model systems for annotation**	Prompting large language model systems for systematic annotation with schema-constrained output (json). The model's intrinsic schema vs. your guideline; responsible use and its limits. Running local open-weight models via LM Studio	Reading: Pagel/Pichler (2026), "Evaluating Humanities Theory Alignment in Large Language Models: Incremental Prompting and Statistical Assessment" Group work: finalize your annotation guidelines, prepare gold annotations for model fine-tuning or few-shot prompting
Session 12 02.06.27	**Data Analysis: Analysis and (multimodal) bias**	Analysis of your data with different approaches; large language model systems as a judge experiments; face analysis or speech recognition; Bias in data, research questions, and results; responsible AI; theory pluralism revisited.	Reading: Haltermann/Keith (2026), "Codebook LLMs: Evaluating LLMs as Measurement Tools for Political Science Concepts" Group work: play around with open-weight large language model systems and compare output to your gold annotations

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08.06.27	**Deadline: Portfolio Part 2**	Submit your second portfolio chapter (operationalization and guidelines for manual annotation) as a PDF on Moodle.	
Session 13 09.06.27	**Poster preparation**	What is a scientific poster and how do I prepare it?	Group work: discuss what you see in your data analysis results, formulate your interpretation and prepare your material as short bullet points for poster preparation
Session 14 16.06.27	**Data interpretation: Answering the research question**	What do you see in your data analysis results? Accompanied writing session: Write up your data analysis and interpretation section with direct feedback	Group work: refine your group poster and prepare your presentation
20.06.27	**Deadline: Poster Submission**	Submit your poster by midnight as a PDF on Moodle.	
Session 15 23.06.27	**Conference simulation**	Group project presentations: poster slam and gallery walk. Conclusion and outlook: what comes next with your MA in Digital Humanities. Feedback and open questions.	Instructional reading: How to ask and answer questions in a conference setting? Group work: prepare the presentation of your poster by equally distributed parts
Session 16 30.06.27	**Conclusion and Outlook**	Summary of the class, reflection and peer feedback; Accompanied writing session: Revise your portfolio parts 1 and 2; Write up a first draft of your reflection chapter for the portfolio part 3: reflection of your personal learning over the semester	Group work: discuss and write down notes from the peer feedback
15.07.27	**Deadline: Portfolio Part 3**	Submit your third portfolio chapter (reflection of your personal learning over the semester) as a PDF on Moodle.	